

Doping Stability Report -- Sn in LiCoO₂ at 350 C

Generated: 2026-06-02 17:38 Host: LiCoO₂ Dopant: Sn

Executive Summary

Sn preferentially occupies the Co layer. The formation energy difference between the two layers is +9.943 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00627 Å²/ps (stable), compared to 0.00134 Å²/ps on the Li site. The Li-site E_f is approximate due to a charge surplus of +3 that cannot be modelled in CHGNet.

Site	E _f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	+2.373	+1	Yes	FAIL	PASS	MIGRATION / UNFAVORABLE
Li	+12.316	+3	Yes	PASS	PASS	METASTABLE

Sn at the Co layer -- MIGRATION / UNFAVORABLE

Charge situation

Sn replaces Co with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E _f	+2.3730 eV	FAIL	< 1.0 eV to pass
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E_f = +2.373 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Sn incorporation is thermodynamically unfavorable under equilibrium conditions at the Co site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	0.00627 Å²/ps	FAIL	< 0.005 to pass (plateau = stable)
Final MSD	0.168 Å²	----	
Max displacement	0.719 Å	----	
Coordination (min/mean/max)	6 / 6.0 / 7	PASS	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.049 Å (relaxed: 1.9)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00627 Å²/ps. The slope exceeds the plateau threshold, suggesting mild drift or site-hopping. Final MSD = 0.168 Å²; maximum displacement from starting position = 0.72 Å.

Coordination fluctuated between 6 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6).

This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.049 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: MIGRATION / UNFAVORABLE

The formation energy is favorable but Sn migrates away from the Co site during MD. The relaxed structure

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is not the true resting site. Inspect the MSD and trajectory to identify where the dopant ends up.

Sn at the Li layer -- METASTABLE

Charge situation

Sn replaces Li with a charge surplus of +3. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	+12.3162 eV	FAIL	< 1.0 eV to pass
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E_f = +12.316 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Sn incorporation is thermodynamically unfavorable under equilibrium conditions at the Li site.

Molecular Dynamics Stability (350 C, 25 ps)

MSD slope (late-time)	0.00134 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.034 Å²	----	
Max displacement	0.871 Å	----	
Coordination (min/mean/max)	5 / 6.0 / 7	PASS	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.093 Å (relaxed: 2.0)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00134 Å²/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.034 Å²; maximum displacement from starting position = 0.87 Å.

Coordination fluctuated between 5 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6).

This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.093 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: METASTABLE

Sn sits stably on the Li site during MD at 350 C, but the formation energy exceeds the 1.0 eV equilibrium threshold. This configuration is metastable -- it may be achievable via non-equilibrium synthesis routes such as ion exchange or rapid quenching.

Site Preference Comparison

The Co site is preferred by 9.943 eV over the Li site. Formation energy: +2.373 eV (Co) vs +12.316 eV (Li).

Sn preferentially occupies the Co layer. The formation energy difference between the two layers is +9.943 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00627 Å²/ps (stable), compared to 0.00134 Å²/ps on the Li site. The Li-site E_f is approximate due to a charge surplus of +3 that cannot be modelled in CHGNet.

Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation.

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Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.